Lecture 1: Introduction to Computability

Models of Computation

https://clegra.github.io/moc/Novi-Sad.html

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University of Novi Sad
Novi Sad, Serbia

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Thanks to

- our hosts:
  - Jovanka Vanja Pantović (Univ. Novi Sad)
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- Alessia Dell'Elce (GSSI administration)
- PNRR-TNE DESK (Developing Shared Knowledge)

- Emilio Tuosto (idea of the visit)
- Dragiža Zunić (help & local expertise)

Course overview

intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	**Three more Models of Computation**
computation and decision problems, from logic to computability, overview of models of computation relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	Post's Correspondence Problem, Interaction-Nets, Fractran
	imperative programming	*algebraic programming*	*functional programming*	

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  - questions where the answer may depend on computation
  - algorithm examples
  - unsolvable problems

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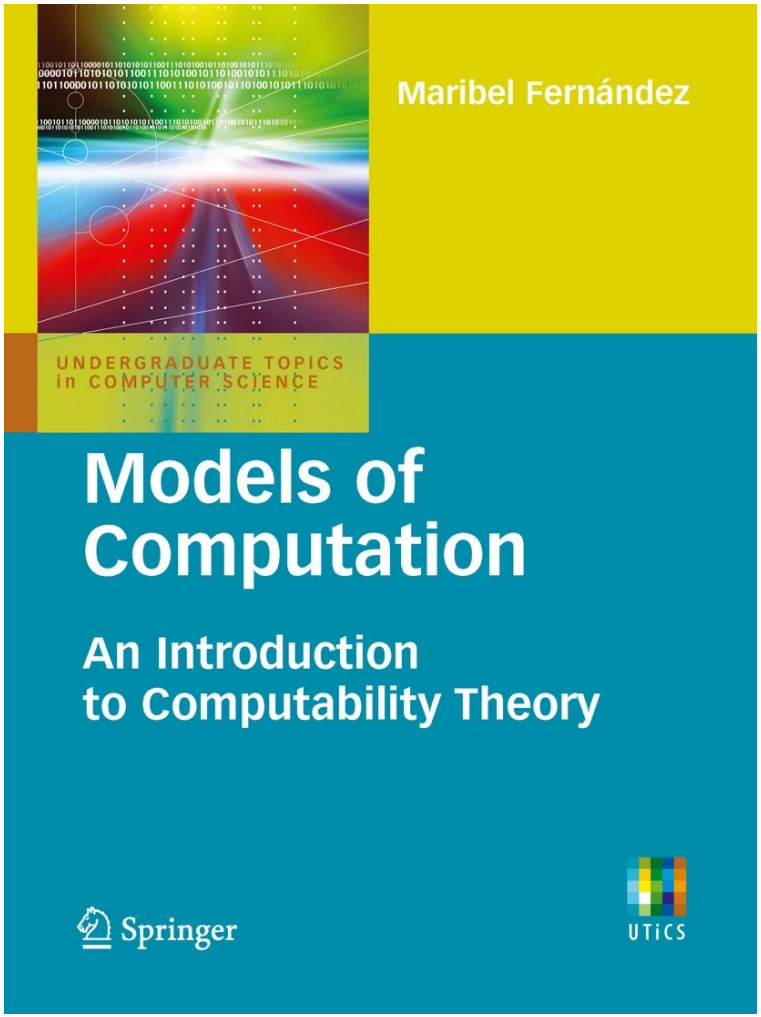
- references

Book


Maribel Fernández
UNDERGRADUATE TOPICS
in COMPUTER SCIENCE
Models of
Computation
An Introduction
to Computability Theory
Springer
UTiCS

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A: Yes, if the truth table for ϕ contains (in the row for ϕ) only "T"; no otherwise.

(Comput.) Yes-or-no-questions / Decision problems

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A decision method for A in E is a method by which, given an element $a \in E$, we can decide in a finite number of steps whether or not $a \in A$.

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The decision problem for A in E is solvable (the set A in E is (effectively) calculable) if there exists a decision method for A in E.

(Comput.) What-questions / Computation Problems

Example

Computing the greatest common divisor
Instance: a pair $\langle a, b \rangle$ of numbers $a, b \in \mathbb{N}$ with $a, b > 0$.
Question: What is $\gcd(a, b)$, the greatest common divisor of a and b?

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Suppose $F : A \to B$ is a mapping, where the elements of A, B are finitely describable objects.

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A mapping F is calculable if there exists a computation method for F.

Representing function

Let $P(a_1, \ldots, a_n)$ be an n-ary number-theoretic predicate.

The representing function f of P:

$$f(a_1, \ldots, a_n) := \begin{cases} 1 & \ldots P(a_1, \ldots, a_n) \text{ is true} \\ 0 & \ldots P(a_1, \ldots, a_n) \text{ is false} \end{cases}$$

Hence:
A decision procedure can be handled as a computation procedure f by taking ‘0’ for ‘yes’, and ‘1’ for ‘no’.

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– Similar for a decision methods.

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Due to $3 \mid 12$ and $(\star)$ we conclude:

A: Yes. (Infinitely many solutions, e.g. $x = 4$ and $y = -8$.)

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Examples (Shoenfield)

- methods that involve chance procedures: tossing a coin

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- methods that involve chance procedures: tossing a coin
- methods involving magic: asking a fortune teller
- methods that require (unformalised, unmechanised) insight

Effectively calculable?

Example

Hilbert's 10th Problem
Instance: An equation $p(x_1, \ldots, x_n) = 0$, where
p a polynomial with integer coefficients.
Question: Is the equation solvable for $x_1, \ldots, x_n \in \mathbb{Z}$?

Instances based on quadratic polynomials are of the form $ax^2 + bxy + cy^2 + dx + ey + f = 0$ with $a, b, c, d, e, f \in \mathbb{Z}$.

Effectively calculable? – No!

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Theorem (Matijasevic, 1970)

Hilbert's 10th Problem is unsolvable.

David Hilbert (1862–1943)

Problem (Entscheidungsproblem, 1928)

Is there a method for deciding, given a formula ϕ of the predicate calculus, whether or not ϕ is a tautology?

Timeline: From logic to computability

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1937	Post: machine model; Church's thesis as 'working hypothesis' Turing: convincing analysis of a 'human computer' leading to the 'Turing machine'

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Questions/Exercises

1. Suppose $P(a, b)$ is a calculable predicate.
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3. Can computation problems for mappings $F : \mathbb{N}^n \to \mathbb{N}^m$ always be represented by decision problems?

Some Models of Computation

machine model	mathematical model	sort
Turing machine Post machine register machine	Combinatory Logic λ-calculus Herbrand–Gödel recursive functions partial-recursive/μ-recursive functions Post canonical system (tag system) Post's Correspondence Problem Markov algorithms Lindenmayer systems	*classical*
	Fractran	*less well known*
cellular automata neural networks	term rewrite systems interaction nets logic-based models of computation concurrency and process algebra ς-calculus evolutionary programming/genetic algorithms	*modern*
abstract state machines		
hypercomputation		*speculative*
quantum computing bio-computing reversible computing		*physics-/biology-inspired*

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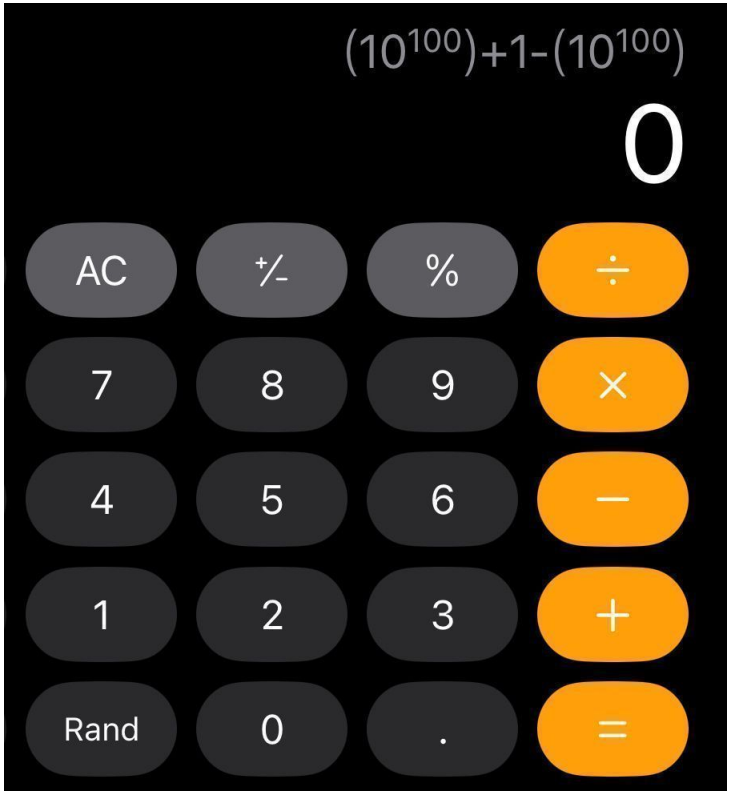
Example MoC relevance: Calculator (1/5)



iOS

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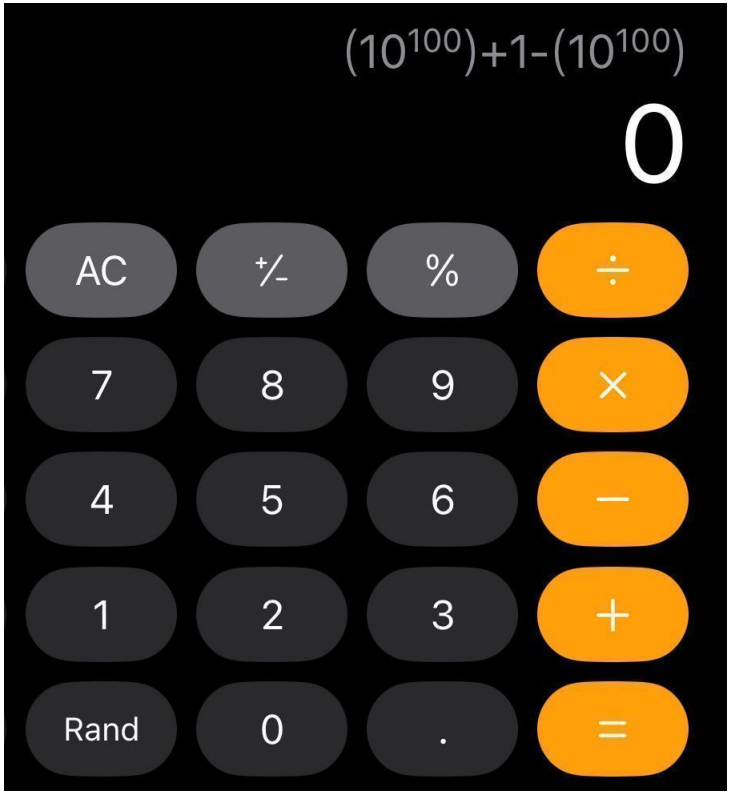

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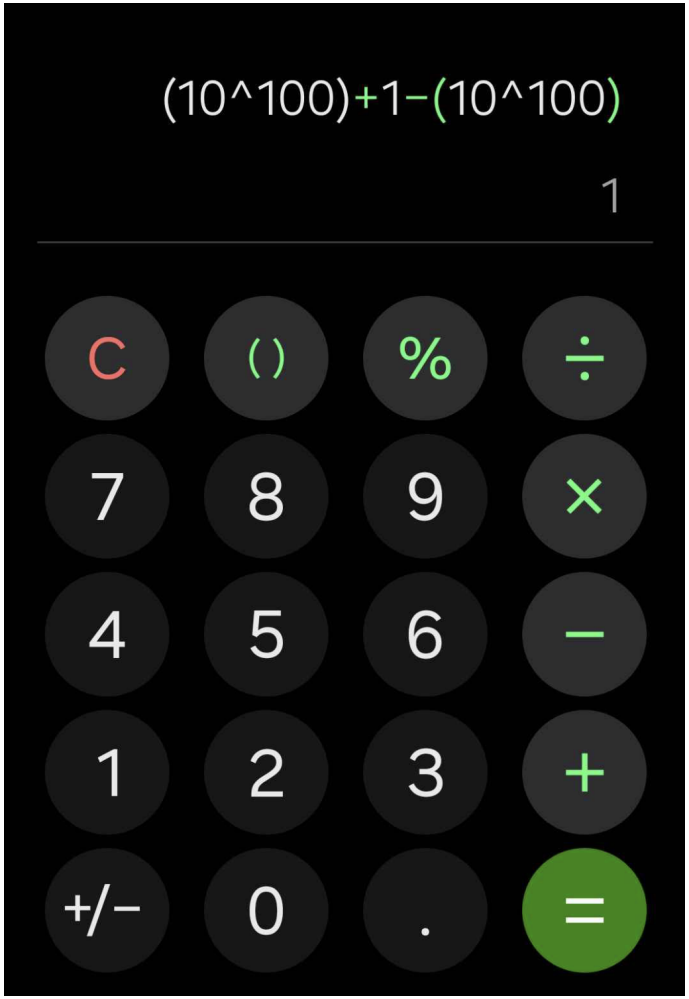
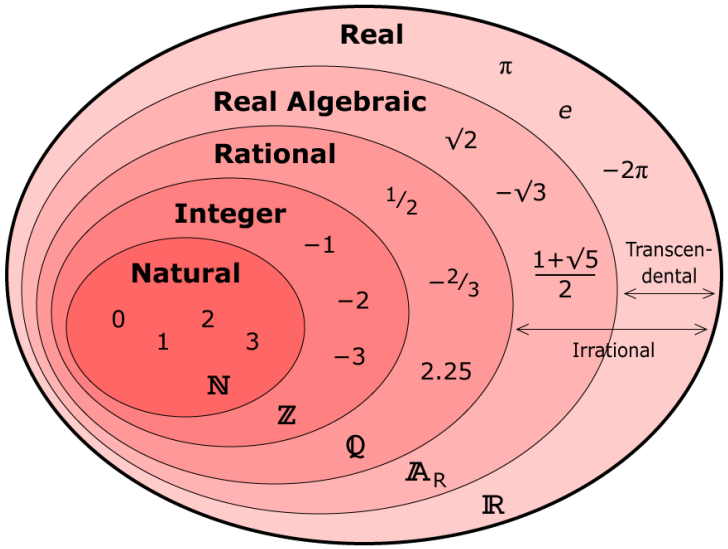


Android

Calculator (2/5): constructive real numbers



subclasses of real numbers $\mathbb{R}$

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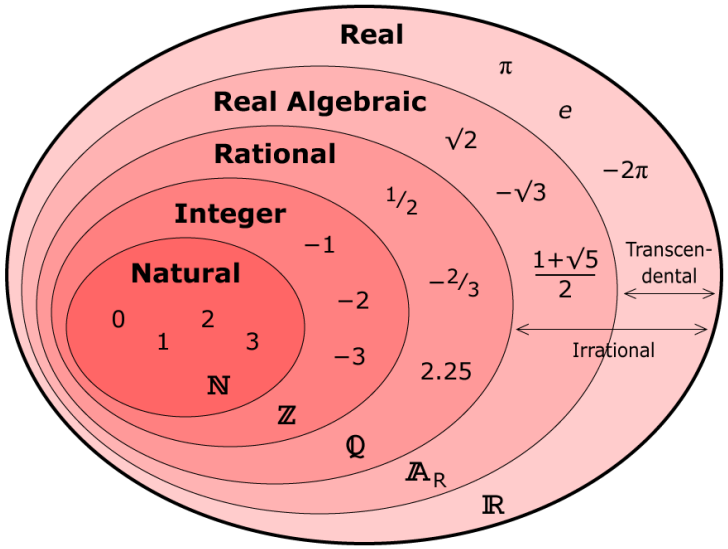

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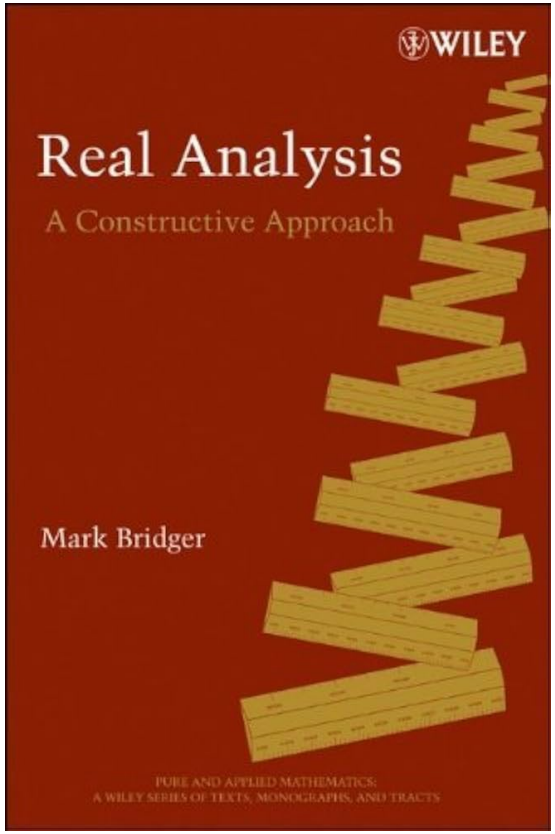

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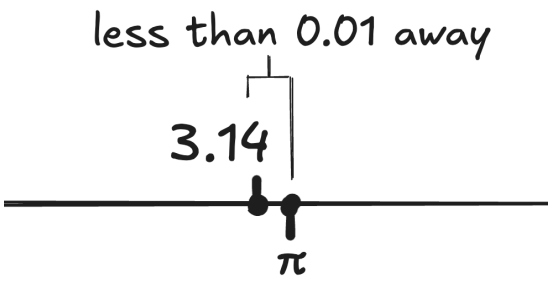
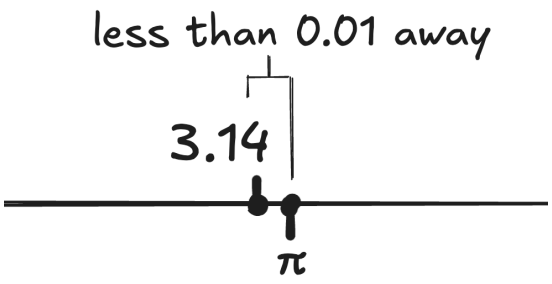


approximating π within 0.01

Calculator (3/5): constructive real numbers



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A real number $x \in \mathbb{R}$ is constructive if:

- there exists a program P_x that for every bound $0 < \delta \in \mathbb{Q}$ returns a rational approximation $P_x(\delta) \in \mathbb{Q}$ of x with $|x - P_x(\delta)| < \delta$.

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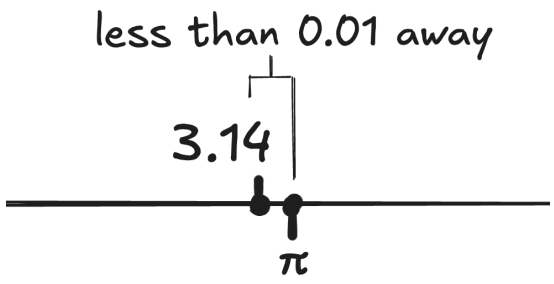

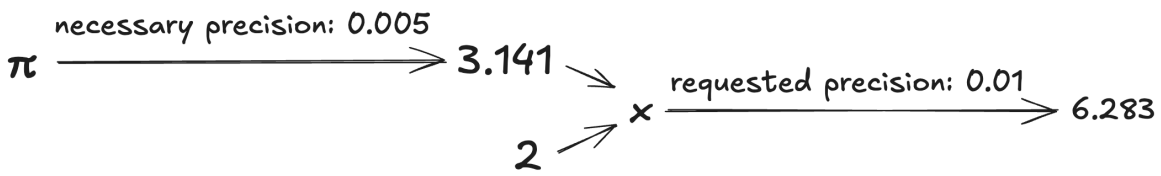

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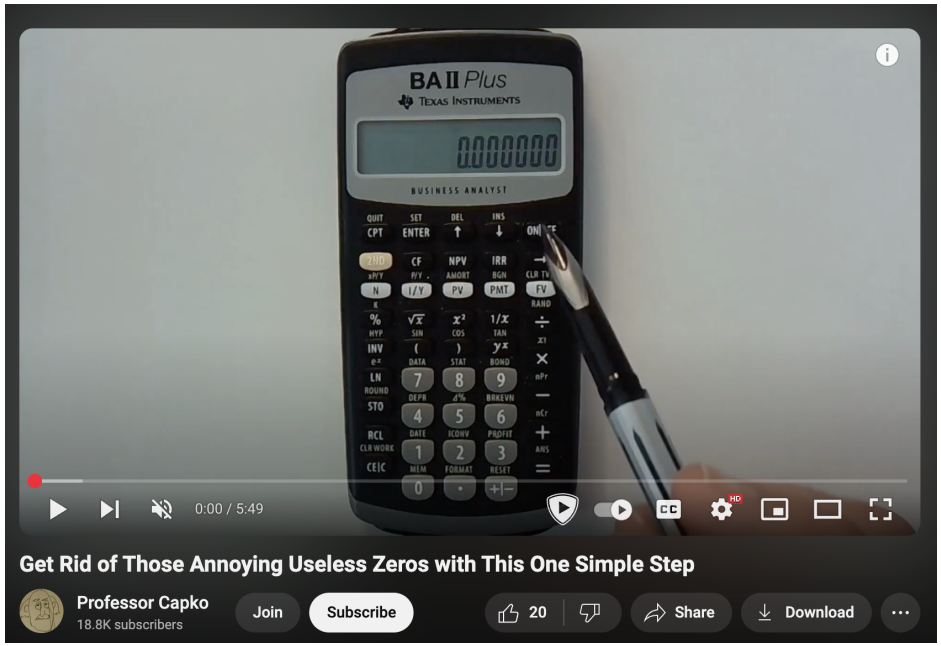

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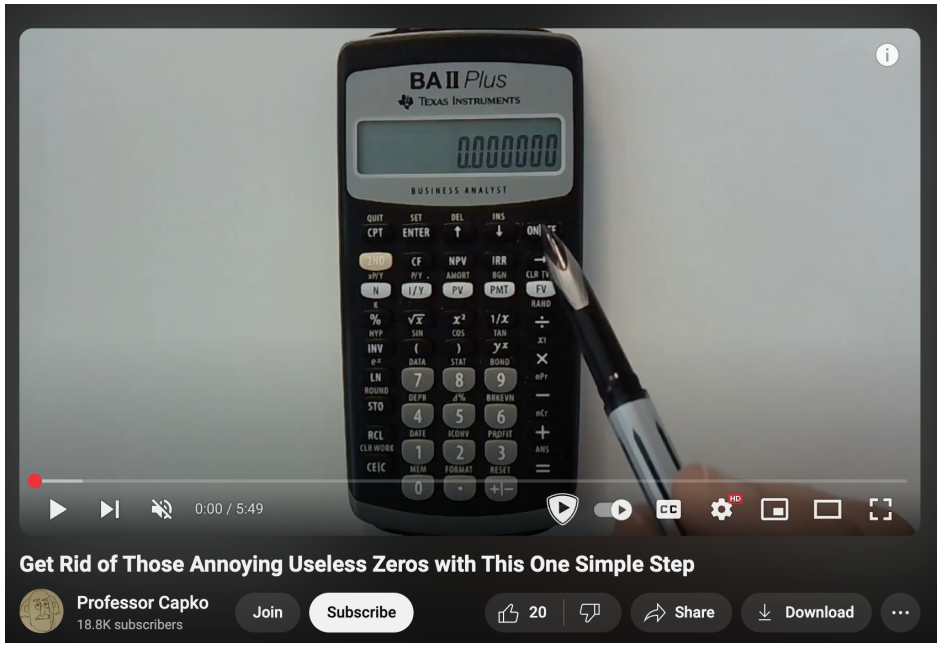

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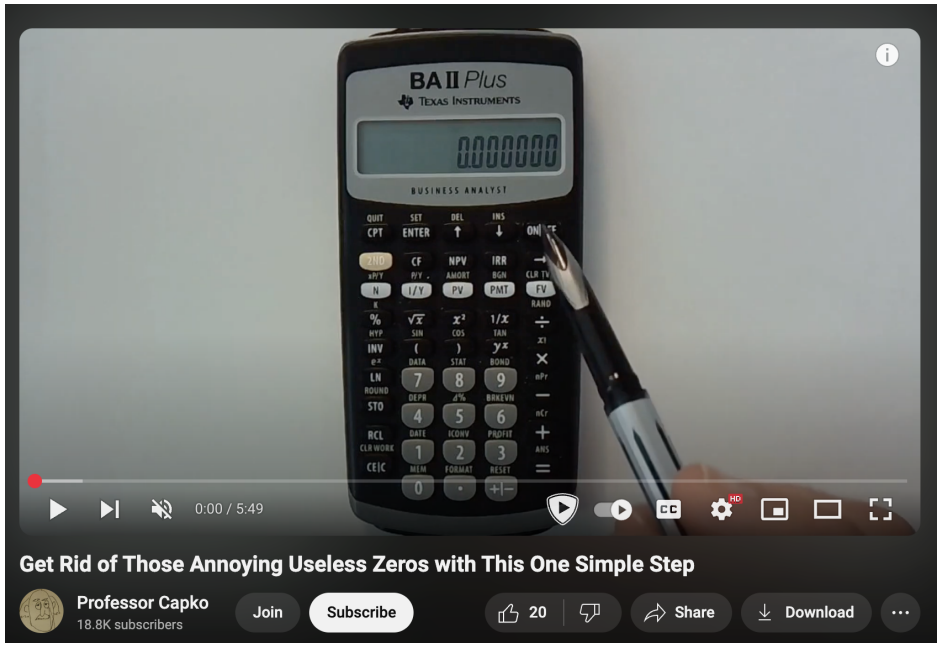

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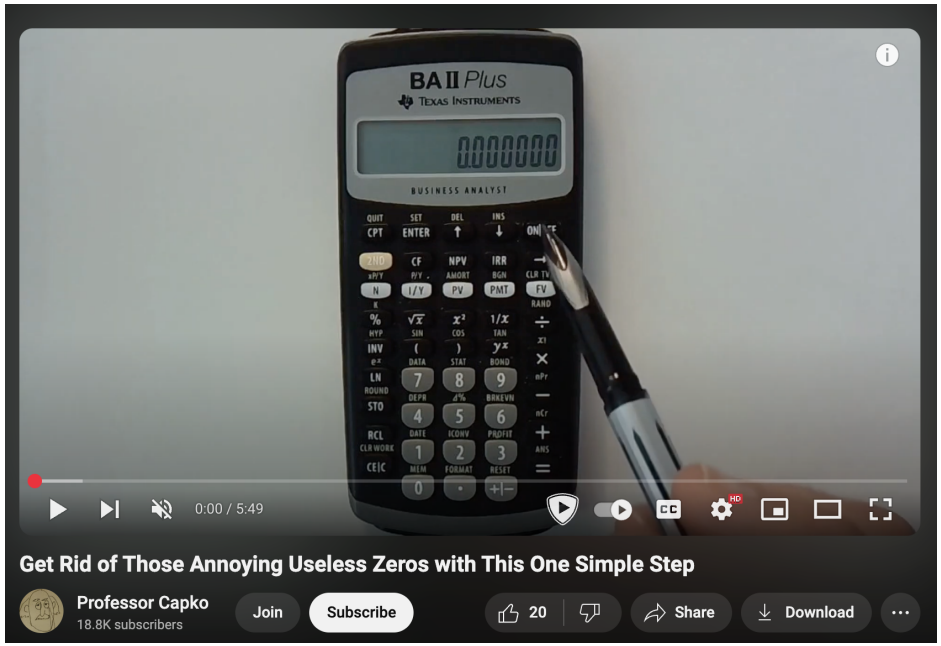

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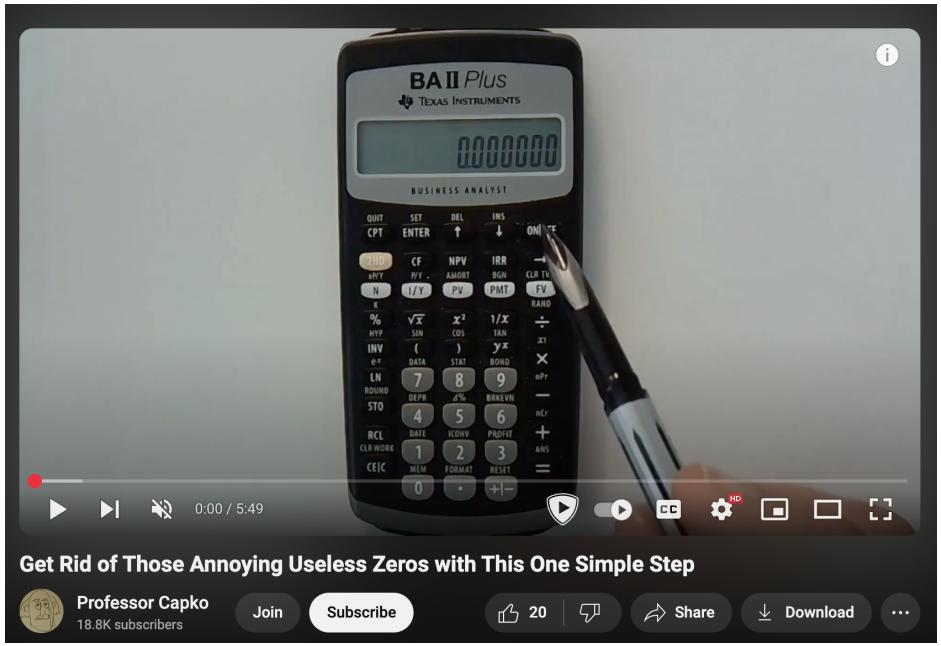

Undecidable problem

Article Talk

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(July 2019) Learn more

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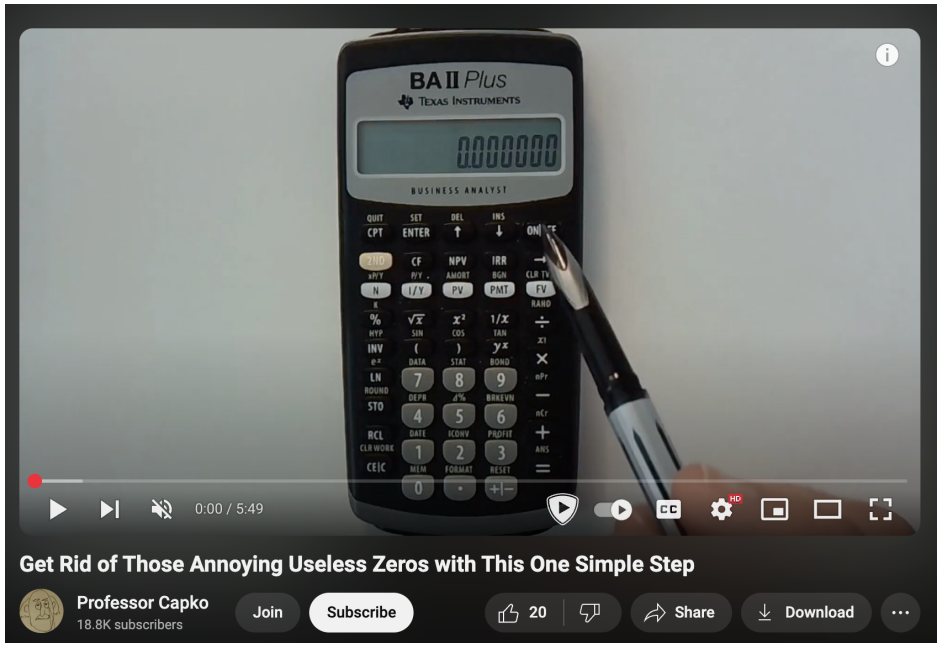

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- Credits: tech-blogger Chad Nauseam (link) for post
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- study the computational complexity of decidable problems (classification of problems into hierarchies)

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Rewriting

- study in a systematic way the operational and denotational aspects of MoC's like λ-calculus, CL, string rewriting, term rewriting, interaction nets

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Linguistics

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Recommended reading

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2. Turing machine motivation: Turing's analysis of a human computer: Part I of Section 9, pp. 249–252 of:
   - Alan M. Turing's: *On computable numbers, with an application to the Entscheidungsproblem*', Proceedings of the London Mathematical Society (1936), [4], http://www.wolframscience.com/prizes/tm23/images/Turing.pdf.

Course overview

intro	*classic models*			*additional models*
Introduction to Computability	**Machine Models**	**Recursive Functions**	**Lambda Calculus**	**Three more Models of Computation**
computation and decision problems, from logic to computability, overview of models of computation, relevance of MoCs	Post Machines, typical features, Turing's analysis of human computers, Turing machines, basic recursion theory	primitive recursive functions, Gödel–Herbrand recursive functions, partial recursive funct's, partial recursive = = Turing-computable, Church's Thesis	λ-terms, β-reduction, λ-definable functions, partial recursive = λ-definable = Turing computable	Post's Correspondence Problem, Interaction-Nets, Fractran
	imperative programming	*algebraic programming*	*functional programming*	

References I

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